

C Revision Map

Condensed core C concepts with quick reference examples for active recall.

1. Basics

Data types, variables, operators, precedence.

```
int a=5, b=2;
printf("%d", a/b); // 2 (integer division)
```

2. Control Flow

if-else, switch, logical evaluation.

```
int x=10;
if(x>5) printf("Yes"); else printf("No");
```

3. Loops

Iteration and control statements.

```
for(int i=0; i<5; i++) printf("%d ", i);
```

4. Functions

Function structure and recursion.

```
int fact(int n){ return (n<=1)?1:n*fact(n-1); }
```

5. Arrays & Strings

Traversal and manipulation.

```
char str[]="abc";
printf("%c", str[1]); // b
```

6. Pointers

Memory address handling and dereferencing.

```
int x=10;
int *p=&x;
printf("%d", *p);
```

7. Structures & Unions

Custom data types.

```
struct S{int a; float b;};  
struct S s1={1,2.5};
```

8. File Handling

Basic file operations.

```
FILE *f=fopen("a.txt", "w");  
fprintf(f, "Hello");  
fclose(f);
```

9. Dynamic Memory

Heap allocation and deallocation.

```
int *p=(int*)malloc(5*sizeof(int));  
free(p);
```